Constructive Method on Goldbach Representations yielding Propagation Bounds and a Prime-Selector Heuristic

Mohamed Amine Belachhab
Fes, Morocco
m.amine.belachhab@gmail.com

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Abstract

In the present paper we formalize a constructive shifting method for propagating Goldbach representations. Starting from a representation $p + q = 2n$, we introduce an algebraic transformation

$$p' = p + 2k, \qquad q' = q - 2k + 2c,$$

so that $p' + q' = 2n + 2c$. Using short-interval prime distribution results, we first prove that the construction generates an infinite family of Goldbach evens. Under the unconditional Hoheisel–Baker–Harman–Pintz exponent $\theta = 0.525$, the number of generated Goldbach evens up to x is at least

$$\Omega\big(x^{0.724375}\big).$$

Under a corrected two-scale Cramér–Granville analysis, the construction propagates with jump parameter

$$c = O\big((\log\log n)^2\big),$$

which gives the conditional generated-density lower bound

$$\Omega\left(\frac{1}{(\log\log x)^2}\right).$$

We also explain why a deterministic max-gap hypothesis alone cannot force full unit-step propagation under the two-scale method, and we formulate a prime-selector heuristic predicting that the unit-step transition $c = 1$ should occur for a density-one set of inputs. The heuristic section is not used as a proof; it is included to identify the exact probabilistic structure that a stronger future theorem would need to control.

Contents

1 Construction Method and Prime Distribution Bounds

Let $(n, p, q, p', q', c) \in (\mathbb{N}^*)^6$ with p, q prime such that

$$p + q = 2n,$$

and let p', q' be primes intended to satisfy

$$p' + q' = 2n + 2c.$$

Let $k \in \mathbb{Z}$. We set

$$p' = p + 2k, \qquad q' = q - 2k + 2c.$$

Then the invariant algebra is

$$p' + q' = (p + 2k) + (q - 2k + 2c) = p + q + 2c = 2n + 2c.$$

Thus the parameter k shifts weight between the two prime coordinates, while the parameter c determines the new even number reached by the construction.

Since p and p' are odd primes for n large enough, the expression $p' = p + 2k$ preserves parity. Similarly, $q' = q - 2k + 2c$ is odd whenever q is odd. Thus parity is consistent throughout the construction.

We use the following short-interval prime input.

Hoheisel–Baker–Harman–Pintz. There exists $\theta \in (1/2, 1)$ such that, for all sufficiently large m, there is a prime in

$$(m,\, m + 2m^{\theta}]$$

and also a prime in

$$[m - 2m^{\theta},\, m).$$

The best currently available unconditional exponent of this form is $\theta = 0.525$ due to Baker, Harman and Pintz. Conditionally under the Riemann Hypothesis, one may take $\theta = 1/2 + \varepsilon$.

Applying the backward interval to

$$X = 2n - 5,$$

there exists a prime

$$p' \in \big(2n - 5 - 2(2n - 5)^{\theta},\, 2n - 5\big).$$

Since $p' = p + 2k$ and $p = 2n - q$, this gives

$$2n - q + 2k \in \big(2n - 5 - 2(2n - 5)^{\theta},\, 2n - 5\big),$$

and hence

$$\frac{q - 5 - 2(2n - 5)^{\theta}}{2} < k < \frac{q - 5}{2}.$$

For integer k, replacing the right endpoint by a harmless closed endpoint gives the admissible range

$$\frac{q - 5 - 2(2n - 5)^{\theta}}{2} < k \le \frac{q - 5}{2}.$$

Lemma 1.1 (Admissible k-interval). *For all sufficiently large n, there exists an integer k in the interval*

$$\left(\frac{q-5-2(2n-5)^{\theta}}{2},\ \frac{q-5}{2}\right]$$

such that

$$p' = p + 2k$$

is prime.

Proof. By the Hoheisel–Baker–Harman–Pintz interval applied below $2n-5$, there exists a prime p' in

$$\big(2n-5-2(2n-5)^{\theta},\ 2n-5\big).$$

Since $p' = p + 2k = 2n - q + 2k$, solving for k gives the stated interval. Because p' and p are odd, $p' - p$ is even, so the resulting value of $k = (p'-p)/2$ is an integer. $\square$

Now define the compressed residual

$$M := q - 2k.$$

Since

$$p' = p + 2k,$$

we have

$$M = q - 2k = q - (p' - p) = p + q - p' = 2n - p'.$$

The above bounds on p' imply

$$5 < M < 5 + 2(2n-5)^{\theta}.$$

We now search for a prime q' above M. By the same short-interval theorem, there exists a prime

$$q' \in (M,\ M + 2M^{\theta}].$$

Writing $q' = M + 2c$ gives

$$0 < c \le M^{\theta}.$$

Lemma 1.2 (Second prime and shift bound). *For all sufficiently large n, the construction produces primes p' and q' such that*

$$p' + q' = 2n + 2c$$

with

$$0 < c \le \big(5 + 2(2n-5)^{\theta}\big)^{\theta}.$$

Proof. The first prime choice gives

$$5 < M < 5 + 2(2n-5)^{\theta}.$$

Applying the short-interval prime result above M, we choose a prime

$$q' \in (M, M + 2M^{\theta}].$$

Since M and q' are odd, $q'-M$ is even, so $q' = M+2c$ for some positive integer c. Moreover,

$$2c = q' - M \le 2M^\theta,$$

and therefore

$$c \le M^\theta < \big(5 + 2(2n-5)^\theta\big)^\theta.$$

Finally,

$$p' + q' = p' + (M + 2c) = p' + (2n - p') + 2c = 2n + 2c.$$

$\square$

Thus the worst-case shift parameter satisfies

$$c_{\max}(n) = \big(5 + 2(2n-5)^\theta\big)^\theta \sim 2^\theta (2n)^{\theta^2}.$$

2 Fundamental Counting and Generated Density Bound

Let $a(x)$ count the number of even integers up to x that are reached by the constructive chain. If the maximum even step between successive constructed Goldbach evens is bounded by

$$\Delta_{\max}(x) = 2c_{\max}(x),$$

then a deterministic covering count gives

$$a(x) \gg \frac{x}{\Delta_{\max}(x)}.$$

In the fixed-exponent case above,

$$\Delta_{\max}(x) \ll x^{\theta^2}.$$

Therefore

$$a(x) \gg x^{1-\theta^2}.$$

Equivalently, the generated density satisfies

$$d_{\text{gen}}(x) := \frac{a(x)}{x/2} \gg x^{-\theta^2}.$$

Theorem 2.1 (Fixed-exponent generated family)**.** *Assume the short-interval prime theorem with exponent $\theta \in (1/2, 1)$. Then the constructive method generates at least*

$$\Omega\big(x^{1-\theta^2}\big)$$

Goldbach evens up to x. In particular, under the unconditional Baker–Harman–Pintz exponent $\theta = 0.525$, it generates at least

$$\Omega\big(x^{0.724375}\big)$$

Goldbach evens up to x.

Proof. From the previous section,

$$c_{\max}(n) \ll n^{\theta^2},$$

so the maximum even step is

$$\Delta_{\max}(n) = 2c_{\max}(n) \ll n^{\theta^2}.$$

Hence the number of constructed even values up to x is at least a constant multiple of

$$\frac{x}{x^{\theta^2}} = x^{1-\theta^2}.$$

For $\theta = 0.525$, one has

$$1 - \theta^2 = 1 - 0.525^2 = 0.724375.$$

$\square$

Remark 2.2. The density obtained here is the density of the even numbers generated by this particular constructive chain. It is not asserted to be the full density of all Goldbach-representable evens.

3 Corrected Two-Scale Cramér–Granville Analysis

The fixed-exponent analysis above may use the same exponent θ at both scales, because the Hoheisel–BHP exponent is uniform for all sufficiently large arguments. The situation is different for a Cramér–Granville type bound. There the natural gap size near x is not x^θ with one fixed exponent, but rather of size $(\log x)^2$. Hence, after the first compression, the second prime search must be carried out at the new compressed scale.

Assumption 3.1 (Cramér–Granville type gap bound)**.** There exists a constant $A > 0$ such that, for all sufficiently large y, there is a prime in

$$(y - A(\log y)^2,\ y)$$

and a prime in

$$(y,\ y + A(\log y)^2].$$

Theorem 3.2 (Two-scale Cramér–Granville propagation)**.** *Assume the Cramér–Granville type gap bound. If $2n$ has a Goldbach representation and n is sufficiently large, then the construction produces a Goldbach representation of*

$$2n + 2c$$

with

$$c = O\big((\log\log n)^2\big).$$

Proof. Set
$$X = 2n - 5.$$
By the backward Cramér–Granville gap bound, choose a prime
$$p' \in (X - A(\log X)^2,\, X).$$
Define
$$M := 2n - p'.$$
Then
$$5 < M < 5 + A(\log(2n-5))^2.$$
Thus
$$M = O((\log n)^2).$$
Now apply the forward Cramér–Granville gap bound at the compressed scale M. There exists a prime
$$q' \in (M,\, M + A(\log M)^2].$$
Since $q' = M + 2c$, we have
$$2c \leq A(\log M)^2.$$
Therefore
$$c \leq \frac{A}{2}(\log M)^2.$$
Because $M = O((\log n)^2)$, we have
$$\log M = O(\log\log n),$$
and hence
$$c = O((\log\log n)^2).$$
Finally,
$$p' + q' = p' + M + 2c = 2n + 2c.$$
$\square$

Corollary 3.3 (Conditional generated density under Cramér–Granville)**.** *Under the Cramér–Granville type gap bound, the construction generates at least*
$$\Omega\left(\frac{x}{(\log\log x)^2}\right)$$
Goldbach evens up to x. Equivalently,
$$d_{\mathrm{gen}}(x) \gg \frac{1}{(\log\log x)^2}.$$

Proof. The previous theorem gives a maximum shift parameter

$$c_{\max}(x) = O((\log \log x)^2).$$

Therefore the maximum even step is

$$\Delta_{\max}(x) = 2c_{\max}(x) = O((\log \log x)^2).$$

The deterministic counting bound gives

$$a(x) \gg \frac{x}{(\log \log x)^2}.$$

Dividing by the number of even integers up to x gives the stated generated density lower bound. □

Remark 3.4. This corrected two-scale result does not imply full Goldbach, nor does it imply generated density 1. It shows instead that the constructive chain has very small conditional gaps: under Cramér–Granville, the jump parameter grows only like $(\log \log n)^2$.

4 Why Max-Gap Control Alone Does Not Force Unit Steps

The preceding section shows that Cramér–Granville gives small jumps, but not necessarily the unit jump $c = 1$. We now state the deterministic obstruction in a clean form.

Let $G(x)$ be a nondecreasing gap function such that primes are guaranteed within distance $G(x)$ above and below sufficiently large numbers of size x.

The first compression gives

$$5 < M < 5 + G(2n - 5).$$

The second prime search gives

$$2c \le G(M) \le G(5 + G(2n - 5)).$$

Thus a deterministic sufficient condition for unit-step propagation is

$$G(5 + G(2n - 5)) < 4.$$

Indeed, if

$$0 < 2c < 4,$$

then the positive integer c must equal 1.

Proposition 4.1 (Two-scale max-gap criterion for unit steps). *A deterministic two-scale max-gap argument forces the transition*

$$2n \longrightarrow 2n + 2$$

only if, eventually,

$$G(5 + G(2n - 5)) < 4.$$

If G is nondecreasing and unbounded, then this condition cannot hold for all sufficiently large n.

Proof. As above, the second prime search gives

$$2c \le G(5 + G(2n - 5)).$$

To force $c = 1$ by a worst-case bound, it is enough to require

$$G(5 + G(2n - 5)) < 4.$$

If G is nondecreasing and unbounded, then

$$5 + G(2n - 5) \to \infty,$$

and therefore

$$G(5 + G(2n - 5)) \to \infty.$$

So the inequality $G(5 + G(2n - 5)) < 4$ cannot hold eventually. $\square$

Remark 4.2. This explains why the full-coverage conclusion cannot be obtained from a growing max-gap hypothesis alone in the corrected two-scale framework. A stronger selector principle is needed: among the available choices of p', one must be able to choose a prime for which the corresponding residual $M = 2n - p'$ also has $M + 2$ prime.

5 A Heuristic Prime-Selector Model for the Unit-Step Transition

The deterministic two-scale analysis proves only

$$c = O\big((\log \log n)^2\big)$$

under a Cramér–Granville type hypothesis. However, this is a worst-case bound. It does not use the fact that there may be many possible choices of the first prime p'. In this section we give a probabilistic heuristic suggesting that, among these possible choices, a unit-step transition

$$2n \longrightarrow 2n + 2$$

should occur very often.

This section is heuristic only. No independence statement strong enough to justify the model is claimed.

5.1 The Selector Count

Let $G(x)$ denote the first prime-search window. In the Cramér–Granville scale, one has

$$G(x) \asymp (\log x)^2.$$

Set

$$N = 2n + 2.$$

The unit-step condition $c = 1$ means

$$p' + q' = 2n + 2 = N.$$

Writing

$$q' = r,$$

this means that both

$$r \qquad \text{and} \qquad N - r$$

are prime, with r lying in the compressed residual range.

Define the unit-step selector count

$$S_G(n) := \sum_{\substack{7 < r \leq 7 + G(2n-5) \\ r \text{ prime}}} \mathbf{1}_{\{N - r \text{ is prime}\}}.$$

Then

$$S_G(n) > 0$$

is exactly the event that the construction admits a unit-step transition.

5.2 Expected Size of the Selector Count

The number of prime candidates $r \leq 7 + G(2n - 5)$ is heuristically

$$\pi(G(2n-5)) \sim \frac{G(2n-5)}{\log G(2n-5)}.$$

For each such prime r, the complementary number $N - r$ has size roughly N, and the heuristic probability that it is prime is about

$$\frac{1}{\log N}.$$

Including the usual local Goldbach correction factor, we expect

$$\mathbb{E}[S_G(n)] \approx \lambda(n),$$

where

$$\boxed{\lambda(n) := \frac{2C_2\, \mathfrak{S}(N)\, G(2n-5)}{\log N \, \log G(2n-5)}}.$$

Here

$$C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right)$$

is the twin-prime constant, and

$$\mathfrak{S}(N) := \prod_{\substack{p \mid N \\ p > 2}} \frac{p-1}{p-2}$$

is the usual singular-series correction for the even integer N.

5.3 Cramér–Granville Scale

Assume now

$$G(x) \asymp (\log x)^2.$$

Let

$$L = \log(2n).$$

Then

$$G(2n-5) \asymp L^2$$

and

$$\log G(2n-5) \sim 2\log L.$$

Therefore

$$\lambda(n) \asymp \frac{2C_2\,\mathfrak{S}(2n+2)\,L^2}{L \cdot 2\log L},$$

so

$$\boxed{\lambda(n) \asymp C_2\,\mathfrak{S}(2n+2)\frac{\log(2n)}{\log\log(2n)}}.$$

In particular,

$$\lambda(n) \to \infty \qquad (n \to \infty).$$

Thus the model predicts that the number of available unit-step selectors grows slowly but unboundedly.

5.4 Poisson Failure Model

We now model $S_G(n)$ as a Poisson-type random variable with mean $\lambda(n)$. This is only a heuristic model, after local congruence obstructions have been absorbed into the singular factor.

Under this model,

$$\mathbb{P}(S_G(n) = 0) \approx e^{-\lambda(n)}.$$

Since $S_G(n) = 0$ is the failure of unit-step propagation, the heuristic failure probability is

$$\boxed{\mathbb{P}(c \neq 1) \approx \exp\left(-\frac{2C_2\,\mathfrak{S}(2n+2)\,G(2n-5)}{\log(2n+2)\ \log G(2n-5)}\right)}.$$

In the Cramér–Granville scale, this becomes

$$\boxed{\mathbb{P}(c \neq 1) \approx \exp\left(-C_2\,\mathfrak{S}(2n+2)\frac{\log(2n)}{\log\log(2n)}\right)}.$$

This tends to 0 as $n \to \infty$. Hence the heuristic predicts that although the deterministic theorem only gives

$$c = O\big((\log\log n)^2\big),$$

the stronger event

$$c = 1$$

should occur for a density-one set of inputs.

5.5 Heuristic Density Consequence

Let

$$F(x) := \#\{n \le x : S_G(n) = 0\}$$

be the number of failed unit-step transitions up to x. The Poisson model suggests

$$F(x) \approx \sum_{n \le x} e^{-\lambda(n)}.$$

Under the Cramér–Granville scale, since

$$\lambda(n) \asymp C_2\, \mathfrak{S}(2n+2) \frac{\log n}{\log\log n},$$

one expects

$$\frac{F(x)}{x} \to 0.$$

Thus the heuristic predicts that the construction propagates by unit steps for almost all n, although this is not proved here.

5.6 Numerical Scale at n = 10ˆ69

As an illustration, take

$$n = 10^{69}.$$

Then

$$\log(2n) \approx 159.57, \qquad \log\log(2n) \approx 5.07.$$

Using the normalized Cramér scale $G(x) \approx (\log x)^2$, the heuristic mean is approximately

$$\lambda(10^{69}) \approx C_2\, \mathfrak{S}(2n+2) \frac{159.57}{5.07}.$$

Since

$$C_2 \approx 0.6601618,$$

we get

$$\lambda(10^{69}) \approx 20.8\, \mathfrak{S}(2n+2).$$

In the average local case $\mathfrak{S}(2n+2) \approx 1$, this gives

$$\lambda(10^{69}) \approx 20.8.$$

The corresponding Poisson failure probability is

$$e^{-20.8} \approx 9.3 \times 10^{-10}.$$

Thus, at this scale, the heuristic strongly predicts the existence of at least one unit-step selector, even though the deterministic two-scale theorem cannot force it.

6 Comparison with Known Exceptional-Set Results

The unconditional exceptional set

$$E(x) = \#\{\text{even } m \le x : m \ne p + q\}$$

has been studied extensively using the circle method and zero-density estimates. Montgomery and Vaughan established

$$E(x) \ll x^{1-\delta}$$

for some $\delta > 0$. Later work made δ explicit, and Pintz obtained

$$E(x) = O(x^{0.72}).$$

Thus the known analytic theory already proves that the actual density of Goldbach-representable even numbers is 1.

The contribution of the present construction is different. It does not improve Pintz's exceptional-set bound. Rather, it gives a direct propagation mechanism and explicit generated-family lower bounds:

$$\Omega(x^{0.724375})$$

unconditionally from BHP, and

$$\Omega\left(\frac{x}{(\log\log x)^2}\right)$$

under Cramér–Granville. The heuristic selector model then suggests why the actual unit-step behavior of the construction may be much stronger than the worst-case deterministic max-gap analysis.

7 Open Problems and Future Directions

The constructive framework formalized in this paper introduces a dynamic shifting machinery that opens several directions for further study.

7.1 Prime-Selector Conjecture

The deterministic two-scale argument cannot force $c = 1$ from a growing max-gap hypothesis alone. A natural replacement is the following selector statement.

Conjecture 7.1 (Residual prime-selector conjecture)**.** For all sufficiently large n, there exists a prime

$$p' \in (2n - 5 - G(2n),\, 2n - 5)$$

such that

$$2n - p' + 2$$

is prime.

This conjecture is exactly the structural condition needed for the construction to propagate by unit steps.

7.2 Second-Moment and Sieve Models for the Selector Count

The heuristic selector count

$$S_G(n) = \sum_{\substack{7<r\le 7+G(2n-5)\\ r \text{ prime}}} \mathbf{1}_{\{2n+2-r \text{ is prime}\}}$$

should be studied beyond expectation. A rigorous second-moment estimate, or a suitable short-interval sieve model with singular-series control, could clarify whether the predicted failure probability

$$e^{-\lambda(n)}$$

can be made rigorous on average.

7.3 Distribution of the Shift Parameter c

The deterministic bounds control only the worst-case size of c. Empirically, or under probabilistic models, the actual required shift may be much smaller. It is therefore natural to study the distribution of the minimal admissible c as a localized arithmetic function of n.

7.4 Application to General Linear Additive Frameworks

The algebraic structure

$$p' = p + 2k, \qquad q' = q - 2k + 2c$$

preserves the parity sum while allowing controlled translation. A possible direction is to ask whether analogous shifting frameworks can be adapted to other additive problems involving primes.

7.5 And much more can be done to improve on this.

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